Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Frantz, Jonathan M Ph.D.		
Project Title:	Using CO2 Gas Exchange to Determine the Factors Controlling Respiration and Carbon use Efficiency in Crop Plants		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	slvyt@cc.usu.edu	**Fax:**	FY 435-797-2605
PI Organization Type:	UNIVERSITY	**Phone:**	435-797-2605
Organization Name:	Utah State University		
PI Address 1:	Plants, Soils, and Biometeorology		
PI Address 2:	Utah State University		
PI Web Page:	www.usu.edu/cpl		
City:	Logan	**State:**	UT
Zip Code:	84322	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2000	**End Date:**	06/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Carbon use efficiency (CUE), the ratio of carbon gained to the total carbon fixed over a 24-h period, is primarily determined by the rate of dark respiration. Determining control of respiration is important to crop productivity. Some studies indicate that respiration is controlled by supply and others indicate it is controlled by demand. Both paradigms cannot be equally true. The goal of this research is to determine the effects that light and temperature have on CUE of crops as well as to construct a model that will predict the CUE for entire life cycles.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Data has been useful in clarifying existing literature, incorporating into crop production models, and providing unique methods to determine growth and maintenance respiration, which is an important component to accurately predicting growth and yield of plants. These studies are unique in that they are the first to show that CUE is influenced consistently by temperature and in quantifying the rate at which plant canopies recover after temperature and shade treatments.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)